Residential House Price Prediction - Data Collection

Project Overview:

Project: Residential House Price Prediction

Goal: Predict house sale prices based on property characteristics such as the number of bedrooms, total rooms, roof style, fireplace availability, garage features, pool area, and other housing attributes.

Learning Type: Supervised Learning

Model Type: Regression

About Dataset:

This dataset contains detailed information about residential properties, including structural features, amenities, garage specifications, exterior characteristics, and sale-related information. The objective is to analyse how different property attributes influence house prices and build a machine learning model capable of accurately predicting the sale price of a house.

The dataset can be used to identify the most influential factors affecting property value and support data-driven decision-making in the real estate industry.

Dataset Features:

- **Bedrooms:** Number of bedrooms in the house.
- **Kitchen:** Number and quality of kitchens.
- **Total Rooms:** Total number of rooms above ground level.
- **Roof Style:** Type of roof design used in the house.
- **Roof Material:** Material used for roofing.
- **Fireplaces:** Number of fireplaces available.
- **Fireplace Quality:** Quality rating of fireplaces.
- **Garage Type:** Type of garage attached to the property.
- **Garage Year Built:** Year the garage was constructed.
- **Garage Finish:** Interior finish level of the garage.
- **Garage Cars:** Number of vehicles the garage can accommodate.
- **Garage Area:** Total garage area in square feet.
- **Pool Area:** Area occupied by the swimming pool.
- **Pool Quality:** Quality rating of the pool.
- **Fence:** Type and quality of fencing around the property.
- **Miscellaneous Features:** Additional property features not covered by other attributes.
- **Month Sold:** Month in which the property was sold.

- **Year Sold:** Year in which the property was sold.
- **Sale Type:** Type of sale transaction.
- **Sale Condition:** Condition under which the property was sold.
- **Sale Price:** Final sale price of the property **(Target Variable).**

Dataset Information:

- **File Type:** CSV (.csv)
- **Dataset Domain:** Real Estate / Housing Market
- **Number of Rows:** 1460
- **Number of Columns:** 81
- **Target Variable:** Sale Price

Objective

Develop a regression-based machine learning model that can accurately predict residential house prices using various property characteristics. The model can assist buyers, sellers, real estate agents, and investors in estimating property values and understanding the factors that contribute most significantly to housing prices.
